

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DIGITAL SIGNAL AND IMAGE PROCESSING (01CT0513)	Aim: imulate Discrete time Sequences	
Experiment No: 09	Date:30-08-26	Enrollment No:92410133069

write a python program to simulate smoothening using

median filters

import cv2

import numpy as np

from google.colab.patches import cv2_imshow

Load the image

image = cv2.imread('content/Untitled.jpg')

Define the size of the median filter kernel (should be an odd number)

kernel_size = 5 # You can adjust the size based on the desired

smoothing level

Apply the Median filter for smoothing

smoothed_image = cv2.medianBlur(image, kernel_size)

Display the original and smoothed images

cv2.imshow(image)

cv2.imshow(smoothed_image)

cv2.waitKey(0)

cv2.destroyAllWindows()

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write a python program to simulate sharpening using spatial

high pass filters

import cv2

import numpy as np

from google.colab.patches import cv2_imshow

Load the image

image = cv2.imread('content/Untitled.jpg')

Apply Gaussian smoothing to reduce noise (optional but recommended)

blurred_image = cv2.GaussianBlur(image, (5, 5), 0)

Create a Laplacian kernel for sharpening

laplacian_kernel = np.array([[0, -1, 0],
[-1, 5, -1],
[0, -1, 0]], dtype=np.float32)

Apply the Laplacian filter for sharpening

sharpened_image = cv2.filter2D(blurred_image, -1,
laplacian_kernel)

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Display the original image, blurred image, and sharpened

image

cv2.imshow(image)

cv2.imshow(blurred_image)

cv2.imshow(sharpened_image)

cv2.waitKey(0)

cv2.destroyAllWindows()

code:

import cv2

import numpy as np

import matplotlib.pyplot as plt

Load image

image = cv2.imread("images.jpeg")

Check if image loaded

if image is None:

print("Image not found")

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else:

Gaussian smoothing

kernel_size = (5, 5)

sigma = 1.5

gaussian_kernel = cv2.getGaussianKernel(5, sigma)

gaussian_kernel = np.outer(gaussian_kernel,

gaussian_kernel)

smoothed_image = cv2.filter2D(image, -1, gaussian_kernel)

Sharpening kernel

sharpening_kernel = np.array([

[-1, -1, -1],

[-1, 9, -1],

[-1, -1, -1]

])

sharpened_image = cv2.filter2D(image, -1,

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sharpening_kernel)

Convert BGR to RGB for matplotlib

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

smoothed_rgb = cv2.cvtColor(smoothed_image,

cv2.COLOR_BGR2RGB)

sharpened_rgb = cv2.cvtColor(sharpened_image,

cv2.COLOR_BGR2RGB)

Display

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)

plt.imshow(image_rgb)

plt.title("Original Image")

plt.axis("off")

plt.subplot(1, 3, 2)

plt.imshow(smoothed_rgb)

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plt.title("Smoothed Image")

plt.axis("off")

plt.subplot(1, 3, 3)

plt.imshow(sharpened_rgb)

plt.title("Sharpened Image")

plt.axis("off")

plt.show()

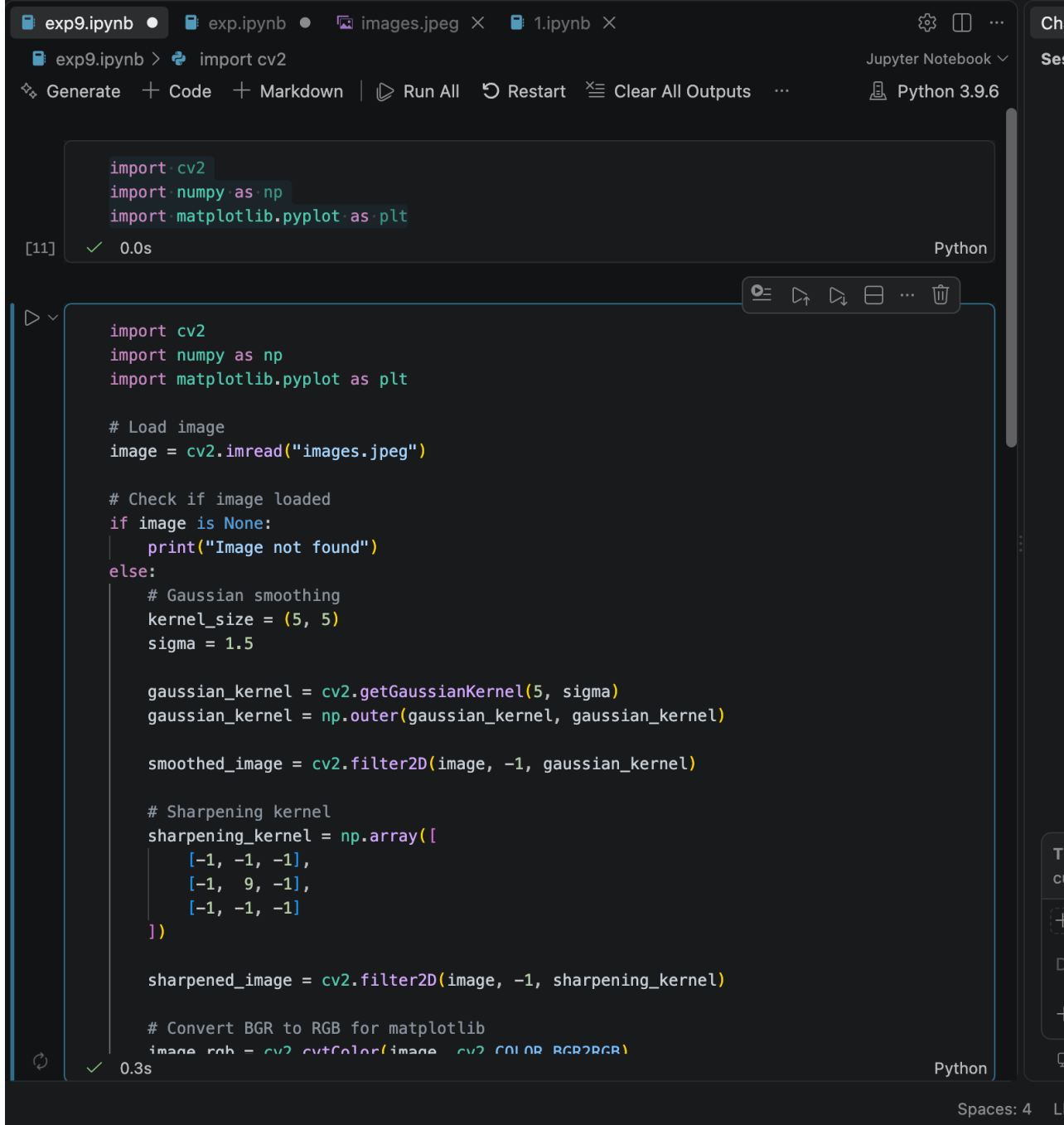
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```
exp9.ipynb • exp.ipynb • images.jpeg × 1.ipynb ×
exp9.ipynb > import cv2
Generate + Code + Markdown | Run All Restart Clear All Outputs Python 3.9.6

import cv2
import numpy as np
import matplotlib.pyplot as plt
[11] ✓ 0.0s Python

import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("images.jpeg")

# Check if image loaded
if image is None:
    print("Image not found")
else:
    # Gaussian smoothing
    kernel_size = (5, 5)
    sigma = 1.5

    gaussian_kernel = cv2.getGaussianKernel(5, sigma)
    gaussian_kernel = np.outer(gaussian_kernel, gaussian_kernel)

    smoothed_image = cv2.filter2D(image, -1, gaussian_kernel)

    # Sharpening kernel
    sharpening_kernel = np.array([
        [-1, -1, -1],
        [-1, 9, -1],
        [-1, -1, -1]
    ])

    sharpened_image = cv2.filter2D(image, -1, sharpening_kernel)

    # Convert BGR to RGB for matplotlib
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    ✓ 0.3s Python

Spaces: 4
```

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```
exp9.ipynb > import cv2

# Convert BGR to RGB for matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
smoothed_rgb = cv2.cvtColor(smoothed_image, cv2.COLOR_BGR2RGB)
sharpened_rgb = cv2.cvtColor(sharpened_image, cv2.COLOR_BGR2RGB)

# Display
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(smoothed_rgb)
plt.title("Smoothed Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image")
plt.axis("off")

plt.show()
```

0.3s Python



Original Image Smoothed Image Sharpened Image

Githublink: <https://github.com/Shashi-kumar-g/DSIP-LAB.git>